Multi-Task Financial NLP: Stance & Sentiment Classification of FOMC Communications and News Headlines

COMP6713 — Natural Language Processing — 2026 T1

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Problem

Two Financial NLP Classification Tasks

We tackle **two complementary 3-class classification problems** over short financial text.

Stance — FOMC Communications

Label the monetary-policy stance of a sentence from U.S. Federal Reserve meetings [1].

- **dovish** — easing bias
- **hawkish** — tightening bias
- **neutral** — no clear bias

"...the Committee decided to lower the target range for the federal funds rate." → dovish

Sentiment — Financial PhraseBank

Label the investor-facing tone of a news headline [2].

- **negative**
- **neutral**
- **positive**

"Operating profit totalled EUR 21.1 mn, up from EUR 18.6 mn a year earlier." → positive

Why It Matters

- **Central-bank stance moves markets.** A single sentence shift between hawkish and dovish language in an FOMC statement can reprice rates, equities, and FX within seconds.
- **News sentiment drives short-horizon prices.** Headline tone correlates with intraday returns and is a staple input to systematic trading and risk models.
- **Both tasks share domain vocabulary** but differ in pragmatics: stance is about *policy direction*, sentiment is about *corporate outlook*. This motivates joint modelling.
- **Labelled data is scarce and expensive.** FOMC: 2,480 sentences; FPB allagree: 2,264 sentences. Techniques that squeeze more signal out of small domain corpora are essential.
- **Practical goal:** a single deployable encoder that does both jobs competitively.

Datasets

Dataset Overview

Name	Sentences	Classes	Label distribution (test)	Notes
FOMC (Trillion Dollar Words) [1]	2,480	3 (dovish / hawkish / neutral)	dovish 130 (26.2%) hawkish 121 (24.4%) neutral 245 (49.4%)	Sentences from FOMC statements, meeting minutes, and press conferences. Neutral-heavy.
Financial PhraseBank *allagree* [2]	2,264	3 (negative / neutral / positive)	negative 61 (13.5%) neutral 278 (61.4%) positive 114 (25.2%)	News headlines with 100% annotator agreement. Highly neutral-skewed; negatives rare.
Loughran–McDonald Lexicon [3]	—	rule-based (pos / neg / uncertainty)	pos ≈110 terms neg ≈230 terms uncertainty ≈90 terms	Domain financial word lists; also includes hawkish (≈40) and dovish (≈50) monetary-policy lists.

- Short texts (mostly < 40 tokens) from the financial domain.
- FOMC and FPB are **imbalanced** with **dominant neutral**; LM Lexicon provides a domain-aware rule-based baseline and auxiliary features.

Preprocessing & Splits

Splits (stratified, `random_state=42`):

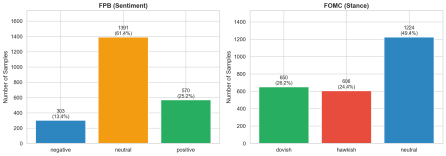
- 70% train / 10% validation / 20% test
- FOMC: 1736 / 248 / 496
- FPB allagree: 1584 / 227 / 453

Text normalisation:

- Unicode NFKC, strip control chars, collapse whitespace
- Lower-case only for TF-IDF baselines
- Keep case and punctuation for BERT-family models (`max_len`=128)

Class weights for FOMC (weighted cross-entropy):

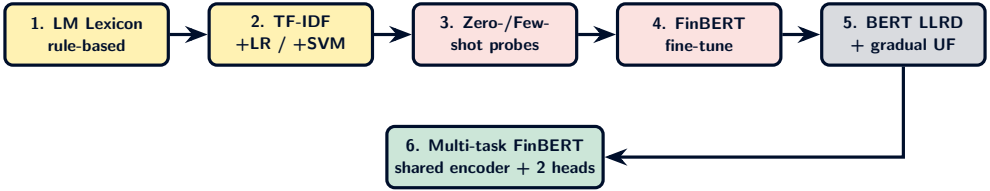
$$w_c = \frac{N}{K \cdot n_c}, \qquad w = (1.272,\ 1.365,\ 0.675)$$



Class distributions for both tasks — neutral dominates in both, negatives and hawkish labels are the minority classes.

Methodology

Pipeline Overview



- Each stage is a **controlled upgrade** over the previous: add features, add domain prior, add fine-tuning, add training recipe, add multi-task structure.
- Every row below is evaluated on the same held-out test sets, so gains are directly comparable.

Baselines: Lexicon and TF-IDF

- **LM Lexicon (rule-based)** [3] — count Loughran–McDonald positive/negative terms per sentence, predict by sign. Domain-specific but brittle.
- **TF-IDF + Logistic Regression** — bigrams (ngram=(1,2)), `max_features`=50k, `sublinear_tf=True`; LR with default ℓ_2.
- **TF-IDF + Linear SVM** — same vectoriser; `LinearSVC`, C=1.0, `class_weight="balanced"` to offset the neutral skew.
- **TF-IDF (trigrams) + LR** — richer context: ngram=(1,3), `max_features`=80k, `min_df`=2.
- **TF-IDF + LM Lexicon (hybrid) + LR** — concatenate TF-IDF vectors with handcrafted counts of LM positive/negative terms.

These establish a *domain-aware but shallow* floor. Any neural model we propose must clearly beat them.

Pre-trained Evaluation: Zero-shot and Few-shot

Zero-shot FinBERT (native 3-way head) [4, 5]

- Use the released {positive, negative, neutral} head directly on FPB.
- For FOMC: proxy map positive $\rightarrow$ hawkish, negative $\rightarrow$ dovish, neutral $\rightarrow$ neutral. Tests whether sentiment is a usable proxy for stance.
- No parameter updates.

Few-shot linear probe (k=16 per class)

- Freeze encoder, use `[CLS]` embedding, fit logistic regression on $3 \times 16 = 48$ examples.
- Compare three frozen backbones:
  - FinBERT (`ProsusAI/finbert`)
  - BERT-base-uncased [6]
  - RoBERTa-base
- Isolates **domain pretraining value** in a tiny-data setting.

Fine-tuning: FinBERT vs. BERT-base LLRD

FinBERT single-task fine-tune

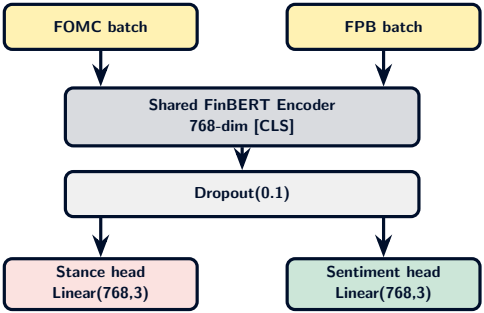
- Backbone: `ProsusAI/finbert`
- Optimiser: AdamW, lr $2e{-5}$, weight decay 0.01
- Schedule: linear warmup (10%) + linear decay
- 5 epochs, batch size 32, `max_len`=128
- Weighted cross-entropy for FOMC
- Strong domain prior, conventional recipe

BERT-base LLRD + Gradual Unfreeze [7, 8]

- Backbone: `bert-base-uncased`
- Layer-wise LR decay: head $2e{-5}$, $\times 0.9$ per depth
- Label smoothing $\varepsilon = 0.1$
- 10 epochs, batch size 32
- Gradual unfreeze: 1 layer group per epoch
- General-purpose backbone, careful recipe

The key comparison: does domain pretraining (FinBERT) beat a better *fine-tuning recipe* on a general backbone (BERT-base)? Results say: **they tie**.

Multi-task FinBERT Architecture



Training details [9]:

- **Alternating batches:** one FOMC batch, then one FPB batch. Each step back-propagates through the shared encoder and the active task's head (the inactive head receives zero gradient).
- AdamW, lr $2e{-}5$, 8 epochs, batch size 32.
- Per-task cross-entropy loss (weighted for stance).
- **Best checkpoint** selected by *average* validation macro-F1 across both tasks.
- Apple M3 Max (MPS), seed 42.

Results

Baseline & Lexicon Results

Model	Sent. Acc	Sent. F1	Stance Acc	Stance F1
LM Lexicon (rule-based) [3]	0.6932	0.5315	0.4153	0.3885
TF-IDF + LR	0.8720	0.8232	0.6089	0.5873
TF-IDF + SVM (LinearSVC)	**0.8940**	**0.8534**	**0.6331**	**0.6061**
TF-IDF (trigrams) + LR	0.8786	0.8310	0.6109	0.5914
TF-IDF + LM Lexicon (hybrid) + LR	0.8543	0.8050	0.6109	0.5863

- **TF-IDF + LinearSVM is the strongest non-neural baseline** on both tasks.
- **Lexicon alone is weak** (FPB F1 only 0.5315, stance F1 0.3885) — sign-counting cannot handle negation or context.
- **Trigrams give only marginal lift** over bigrams (+0.66 pp sentiment, +0.20 pp stance accuracy) — vocabulary grows faster than signal.
- Adding LM-lexicon features to TF-IDF does *not* help — the TF-IDF vocabulary already covers those words.

Pre-trained Models: Zero-shot and Few-shot

Model	Sent. Acc	Sent. F1	Stance Acc	Stance F1
FinBERT zero-shot (native head) [4]	0.9735	0.9650	0.4980	0.4874
FinBERT few-shot (k=16 probe)	**0.9779**	**0.9670**	0.4859	0.4534
BERT-base few-shot (k=16) [6]	0.7417	0.6500	0.3851	0.3744
RoBERTa-base few-shot (k=16)	0.7682	0.6722	0.3730	0.3600

- **FinBERT zero-shot is already near-optimal on sentiment** (F1 0.9650) — its pretraining matches FPB almost exactly.
- **FinBERT dominates both probes**: +30 pp sentiment F1 over BERT/RoBERTa at k=16 — domain pretraining, not size, is what matters in low-data regimes.
- **Stance is much harder zero-shot**: the positive→hawkish / negative→dovish proxy map only reaches F1 0.4874 — stance $\neq$ sentiment.
- The probe tops sentiment but *loses* a little stance F1 vs. native head — 48 examples is too few to learn stance.

Fine-tuned Single-Task Models

Model	Sent. Acc	Sent. F1	Stance Acc	Stance F1
FinBERT (fine-tuned)	0.9669	0.9459	0.6371	0.6194
BERT-base LLRD + Gradual UF [7, 8]	**0.9757**	**0.9670**	**0.6512**	**0.6383**

- **BERT-base with LLRD + gradual unfreeze beats single-task FinBERT on both tasks**: +2.11 pp sentiment F1 (0.9459 → 0.9670) and +1.89 pp stance F1 (0.6194 → 0.6383); sentiment-F1 ties FinBERT few-shot's 0.9670.
- **Takeaway:** a disciplined fine-tuning recipe on a general-purpose backbone can close the domain-pretraining gap [8].
- This is the first evidence in our study that "domain BERT" is not the only path to strong finance-domain results.

Multi-task FinBERT — Headline Result

	Sentiment	Stance
Accuracy	**0.9779**	**0.6492**
Macro-F1	**0.9666**	**0.6384**
Δ vs. single-task FinBERT	+2.07 pp F1	+1.90 pp F1

- One shared encoder, two heads, alternating batches — **improves both tasks** over single-task FinBERT [9].
- Sentiment F1 reaches 0.9666 (matching BERT LLRD's 0.9670); stance F1 reaches 0.6384 (within 0.0001 of BERT LLRD's 0.6383).
- **Free lunch:** no new data, no new parameters beyond a second 768×3 head, and training cost is comparable to single-task.

Results Summary — All 12 Systems

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BERT-base LLRD + Gradual UF	0.9757	**0.9670**	**0.6512**	**0.6383 [BEST]**
Multi-task FinBERT	**0.9779**	**0.9666**	0.6492	**0.6384 [BEST]**

Statistical tie at the top: Multi-task FinBERT and BERT-base LLRD are **indistinguishable** within ±0.1 pp on both macro-F1s. Two very different routes (multi-task + domain backbone vs. single-task + careful recipe) reach the same ceiling.

Error Analysis — Multi-task FinBERT

Stance (FOMC test, N=496)

Class	Per-class F1
dovish	0.6174
hawkish	0.5869
neutral	**0.7109**

- Most errors sit on the **neutral vs. leaning** boundary — hedged language ("*appropriate to maintain*") is genuinely ambiguous.
- Hawkish is the weakest class (lowest count in train and most lexically diverse).

Sentiment (FPB test, N=453)

Class	Per-class F1
negative	0.9500
neutral	**0.9928**
positive	0.9569

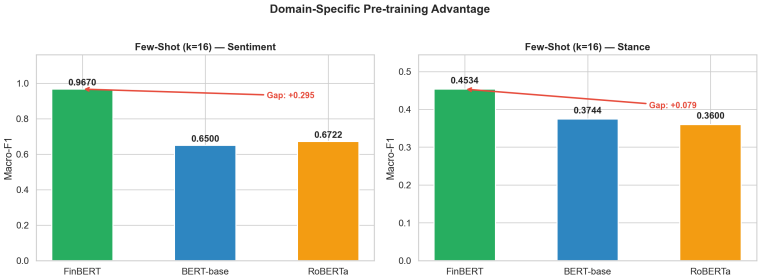
- Only ~ 10 **errors total** across 453 test sentences — sentiment is near-saturated.
- Remaining errors are subtle *comparative* constructions ("*profit fell less than expected*").

Analysis

Key Findings

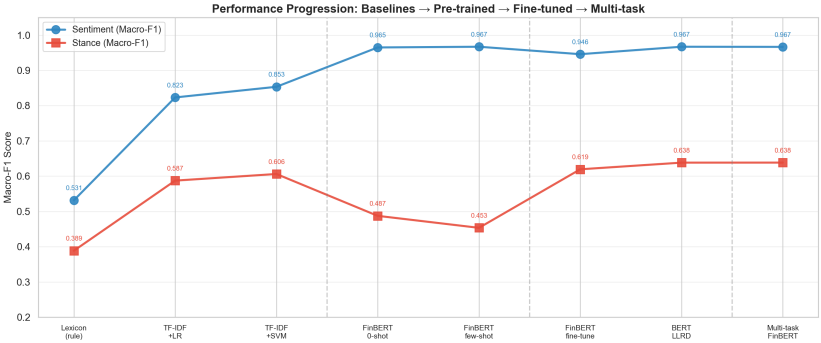
1. **Stance is a fundamentally harder task than sentiment.** Every model loses ≈ 30 pp macro-F1 going from FPB to FOMC. The gap is not about data size — the two datasets are similar — but about *label pragmatics*.
2. **Sentiment is saturated by domain pretraining alone.** FinBERT zero-shot already reaches F1 0.9650; fine-tuning adds almost nothing on FPB. Extra capacity is spent on stance.
3. **A careful fine-tuning recipe rivals domain pretraining.** BERT-base with LLRD + gradual unfreeze + label smoothing *beats* single-task FinBERT on both sentiment (+2.11 pp) and stance (+1.89 pp) [8, 7].
4. **Multi-task training is a free lunch.** One shared encoder + two heads improves single-task FinBERT by +2.07 pp sentiment F1 and +1.90 pp stance F1 [9], and ties BERT LLRD on both.
5. **Zero-shot sentiment $\neq$ stance.** The native FinBERT positive/negative head maps poorly to hawkish/dovish (F1 0.4874) — a reminder that tone and policy direction are different signals.

Domain-Pretraining Gap (k=16 probe)



In the low-data regime, **FinBERT outperforms BERT-base and RoBERTa-base by** ≈ 30 **pp sentiment F1**. Fine-tuning closes part of this gap — but *only* when paired with a careful recipe (LLRD + gradual unfreeze).

Performance Progression Across Methods



The sentiment curve **plateaus** early (FinBERT zero-shot is already near-ceiling); the stance curve keeps climbing through fine-tuning and multi-task. This is where the remaining engineering headroom lies.

Deployment

Deployment — CLI, Demo, and Hub Release

Command-line interface

- `python cli.py --text "The Committee decided to raise the target range."`
- Returns predicted stance and sentiment with class probabilities.
- Loads the multi-task FinBERT checkpoint by default; single-task checkpoints available via `--model`.

Interactive demo

- Gradio app at `http://localhost:7860`
- Text box + live predictions + per-class probability bars.
- Same checkpoint as the CLI — reproducible.

HuggingFace Hub release

Five public model repos under the `Louisnguyen/` namespace:

- `Louisnguyen/finbert-financial-stance`
- `Louisnguyen/finbert-financial-sentiment`
- `Louisnguyen/bert-llrd-financial-stance`
- `Louisnguyen/bert-llrd-financial-sentiment`
- `Louisnguyen/multitask-finbert-financial`

Each repo ships weights, tokenizer config, and an inference snippet.

Summary

Summary

- **Problem:** joint stance (FOMC) and sentiment (FPB) classification on small, imbalanced financial corpora.
- **Approach:** six-stage progression from LM-lexicon rules through TF-IDF, zero/few-shot probes, single-task fine-tuning (FinBERT and BERT-LLRD), and finally multi-task FinBERT with a shared encoder and two heads.
- **Headline result:** multi-task FinBERT and BERT-base LLRD *tie* for best on both tasks (sentiment F1 $\approx$ 0.967; stance F1 $\approx$ 0.638). Multi-task beats single-task FinBERT by +2.07 pp sentiment F1 / +1.90 pp stance F1.
- **Lesson:** domain pretraining and careful fine-tuning recipes are **substitutes**, not complements — either one reaches the ceiling for this data size.

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[3] Tim Loughran and Bill McDonald. **"When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks".** In: *The Journal of Finance* 66.1 (2011), pp. 35–65.

[4] Dogu Araci. **"FinBERT: Financial Sentiment Analysis with Pre-trained Language Models".** In: *arXiv preprint arXiv:1908.10063* (2019).

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